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Physics of the krypton fluoride laser

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The extremely complex kinetics of pumping, quenching, and absorption in krypton fluoride excimer lasers is now relatively well understood. A single comprehensive kinetic model, solved approximately by analytic means and more exactly by a computer code of moderate size, is capable of simulating with great accuracy the performance of KrF lasers pumped by e-beam sustained discharge, and self-sustained discharge. We present the results of applications of this model to interpret and predict the limiting behavior and scaling of KrF lasers. A measure of remaining uncertainties is discussed.

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I. INTRODUCTION

The reports of the bound-free emission of krypton fluoride by Golde,[1] Velazco and Setser,[2] and Brau and Ewing,[3] and the subsequent achievement of laser action on the $^2\Sigma^+_{1/2} \rightarrow {}^2\Sigma^+_{1/2}$ band at 248 nm by Ewing and Brau[4] have triggered an extremely active area of research. The demonstration of a pulsed ultraviolet laser with an intrinsic efficiency of approximately 10% and with both high-energy and high-power applications has stimulated intensive research on the discharge, kinetic, and optical properties of the KrF laser. The dominant processes involved in these areas, interrelated in a complex manner, are now relatively well understood, although some detailed basic research remains to be accomplished.

The KrF emission spectrum shows multiple transitions from a bound electronic upper state to a dissociative lower state.[1-5] Strong emission is observed in fluorescence with a peak near 248 nm, and weaker peaks are observed at shorter wavelengths. The weak undulatory structure arises from the multiple overlap of each upper-state vibrational eigenfunction with that of the dissociative lower state. A cleft in the strong transition is observed at 248.9 nm under some, but not all, pumping conditions. When KrF is pumped intensely, as under lasing conditions, the fluorescence spectrum exhibits a strong peak at 248.4 nm and a weaker peak at 249.1 nm. Tellinghuisen *et al.*[6] tentatively ascribe the peak to a stimulated emission contribution to the spontaneous spectrum from the $v = 0$ and $v = 1$ vibrational states. Only the two peaks at 248.4 and 249.1 nm are observed in the laser spectrum.

The electronic potential curves of KrF have been studied experimentally[6] and theoretically.[7] The covalent lower state is dissociative and correlates with the $Kr(^1S) + F(^2P)$ separated atom limit. The upper state correlates with the $Kr^+(^2P) + F^-(^1S)$ separated ion limit. Strong spin-orbit coupling splits the nearly degenerate $^2\Sigma^+$ and $^2\Pi$ curves of the lowest-lying excited electronic states into three curves.

The close proximity of these curves, especially those of the $C^{1/2}$ and $B^{3/2}$ states, could lead to reduced upper laser level populations if the plasma electrons provide sufficient mixing of states. This possibility is mentioned by Greene and Brau.[8]

The ionic potential energy curve of the upper state crosses the curves of several covalent states at large internuclear separations. This is of central importance to the efficiency of KrF as a laser molecule because it permits several entrance channels to the upper laser level. In particular, KrF can be formed from ions in the reaction

$$Kr^+ + F^- + M \rightarrow KrF^* + M$$

as well as from excited atomic states of krypton by the charge-transfer (harpoon) reactions[62]

$$Kr^* + F_2 \rightarrow KrF^* + F,$$

$$Kr^{**} + F_2 \rightarrow KrF^* + F.$$

In the reactions above, M is a nonreactive collision partner and Kr* and Kr** represent the $5s$ and $5p$ states of krypton, respectively. These krypton excited states store energy effectively because their transitions are either metastable or optically trapped as resonance radiation.

Argon is conventionally used as a background gas in KrF laser mixtures because it stops energetic beam electrons well, is inexpensive compared with krypton, and collisionally transfers its energy to krypton efficiently.[9] The lower-lying argon states also store energy effectively, an additional practical advantage in using Ar/Kr mixtures. The most effective fluorine donor has been F_2, although NF_3 has also been used successfully.[10,11] Since F_2 absorbs KrF radiation and NF_3 does not, NF_3 has an apparent advantage, but the charge transfer of Kr^+ to NF_3 to form NF_3^+ [12] represents a loss to the ion channel of KrF* formation. For this reason, higher fluorescence efficiencies have been observed with F_2 than with NF_3 as the fluorine donor.[13] Mixtures of $Ar/Kr/F_2$ have become standard for e-beam-pumped KrF lasers.

Several pumping schemes have been used for KrF. Krypton fluoride has been pumped directly by electron beams,[4,11,14,15] and KrF lasers have been operated in a discharge mode.[10,16] The discharges have been electron beam

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sustained and self-sustained. The self-sustained discharges have been preionized by electron beams and by ultraviolet radiation.[8] We restrict our discussions to electron-beam-initiated KrF lasers.

Selection of the method used to pump KrF involves consideration of intrinsic laser efficiency, electron-beam energy utility, and discharge stability. Each of these phenomena imposes unique limits on the laser's performance. Intrinsic efficiency,[69] the ratio of laser energy out of the oscillator to energy deposited in the laser mode volume, has an easily determined theoretical upper bound, but in practice it is limited by photoabsorption, quenching, and metastable species ionization.

Direct electron-beam pumping produces an electron-ion pair with an investment of approximately 26.2 eV in the argon-rich gas mixtures typical for KrF experiments.[17,66] Lorents's study of the partitioning of this energy loss among ionization, secondary electron production, and electronic excitation in rare gases allows us to estimate the theoretical intrinsic efficiency upper limit for the direct-pumped KrF laser as 22%. Since the KrF photon represents 5 eV of energy and the Kr* excitation threshold is 9.9 eV, the application of an electric discharge might raise the intrinsic efficiency to as high as 51%. Actually, of course, these figures will be lower in real experiments.

The attainable efficiency of a discharge-pumped laser is strongly related to the discharge enhancement, the ratio of the energy deposited by the discharge to the energy deposited by the electron beam; high values of discharge enhancement ($\geqslant 5$) are needed to counterbalance energy losses due to ionization of metastable species. Discharge enhancement is not the only important consideration, however. All of the beam energy can be used in direct pumping, whereas its energy is only partially deposited in discharge-pumped lasers because the plasma must be kept homogeneous to avoid discharge arcing.[13]

In this paper we present a laser model that incorporates the dominant physical processes in KrF lasers pumped by both electron beams and electric fields. These processes have been identified through extensive simulations of experiments performed at Maxwell Laboratories, Stanford Research Institute, AVCO Everett Research Laboratory, and Northrop Corporation. The experiments cover a wide range and combination of electron-beam currents and sustaining discharge voltages. The simulations predict the experiments with a precision that is remarkable when one considers the idealization of the model.

The kinetic model is presented in Sec. II. There we discuss the multitude of pumping and quenching channels involved in forming and destroying KrF molecules. The paths to KrF formation comprise both direct and tortuous; some quenching is saturable, some is not. The critical role of photoabsorption is discussed qualitatively. Its limitation at high electron densities is demonstrated quantitatively in Sec. IV.

A discussion of the discharge plasma is presented in Sec. III. In particular, the effect of the electron beam on ionization and direct pumping is discussed. An analytic representation is made of the experimentally observed mechanism of discharge current runaway.[18] We interpret the runaway as caused primarily by the absence of an attachment-dominated discharge equilibrium, rather than by the ionization instability proposed by Daugherty *et al.*[19] The dependence of two distinguishable equilibria on electron-beam current and electric field strength is derived, and their stability against density fluctuations is determined.

Some representative laser simulations are presented in Sec. IV. We discuss both direct-pumping and discharge-pumping results. We conclude that inherent discharge losses prevent the discharge-pumped lasers from achieving their theoretical advantage over direct-pumped lasers. General efficiency limits for KrF lasers are described, summarizing both theory and experiment. The optical extraction losses are shown to reduce the intrinsic efficiency considerably below the relatively high ($\sim 22\%$) KrF discharge production efficiency at large electron densities. It is demonstrated that one might theoretically construct a KrF laser with an output of as high as 30 J/liter. But the proposed device would operate at a severe penalty in efficiency and would require operation in a very limited region of parameter space where the e-beam current and discharge voltage must be carefully tailored to unreasonably stringent conditions.

The first physical optics calculations performed for a KrF laser are presented in Sec. V. They are included here because our purpose is to analyze a complete KrF oscillator, not just the discharge plasma and reaction kinetics as previous works have done. We enumerate several problems that need to be addressed. Some of these problems are purely optical, others are related to the interaction between the radiation field and the gain medium inside an optical resonator. Our attention is restricted to one of these problems. We discuss mode buildup time in a positive branch confocal resonator, then indicate numerical problems peculiar to models of lasers with fast reaction kinetics.

II. KINETICS

A. Pumping and quenching

The ionic bonding of the upper level of KrF is fundamentally important to its efficiency as a lasing molecule. The excimer is formed by not only the association of Kr^+ and F^-, but also the curve crossing from covalent states to ionic states at large internuclear distances. In addition, the unbound lower laser level permits intense short laser pulses without the hazard of thermal bottlenecking. Thus, we can characterize KrF as a laser molecule with several entrance channels to the upper laser level combined with a negligible population in the lower laser level. It is, in essence, a one-level laser.

This variety of reaction channels resulting in population of the upper laser level is one of the factors which makes all rare gas–halide kinetic systems extremely complex (especially by the standards of, say, CO_2-N_2 systems). In practice, it is reasonable to assume that all these reactions lead directly to the upper laser level. The questions then become: Which channels dominate under different pumping conditions? What are the inherent limitations of lasers designed to exploit the various channels?

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A set of chemical reactions significant for the KrF laser has been compiled by the authors for use in simulation codes (see Sec. IV). These reactions are displayed in Table I, and henceforward chemical reaction numbers referred to will be those in Table I. Note that for electron-temperature dependent reactions, typical values of these rates for discharge operation (at 2 kV/cm/atm) are given.

Relatively early in the research, it became apparent that the primary distinction between KrF pumping mechanisms is whether the energy reaches the lasing molecule through neutral or ionic species. Although eight reactions produce significant amounts of KrF*, three of these dominate all practical cases:

(1) $Kr^* + F_2 \rightarrow KrF^* + F$ (harpooning),

TABLE I. KrF reaction rate system.

Pumping, quenching and stability reactions

No.	Reaction	Rate constant	Reference
1	$Kr^* + F_2 \rightarrow KrF^* + F$	8.1×10^{-10} cm^3/sec	33
2	$ArF^* + Kr \rightarrow KrF^* + Ar$	3.0×10^{-10} cm^3/sec	32,33
3	$Kr^+ + F^- \rightarrow KrF^*$	1.0×10^{-6} [b] cm^3/sec	32,44
4	$Ar^* + F_2 \rightarrow ArF^* + F$	8.5×10^{-10} cm^3/sec	33
5	$Ar^+ + F^- \rightarrow ArF^*$	1.0×10^{-6} [b] cm^3/sec	44
6	$Kr^{**} + F_2 \rightarrow KrF^* + F$	8.1×10^{-10} cm^3/sec	33
7	$Ar^{**} + F_2 \rightarrow ArF^* + F$	8.5×10^{-10} cm^3/sec	33
8	$Kr_2^+ + F^- \rightarrow KrF^* + Kr$	1.0×10^{-6} cm^3/sec	44,32,40
9	$Ar_2^+ + F^- \rightarrow ArF^* + Ar$	1.0×10^{-6} cm^3/sec	44,32,40
10	$ArKr^+ + F^- \rightarrow KrF^* + Ar$	1.0×10^{-6} cm^3/sec	Estimated
11	$ArKr^* + F_2 \rightarrow KrF^* + Ar + F$	6.0×10^{-10} cm^3/sec	33
12	$KrF^* + Kr \rightarrow 2Kr + F$	2.0×10^{-11} cm^3/sec	33
13	$KrF^* + Ar \rightarrow Kr + Ar + F$	0.0 cm^3/sec	Estimated
14	$KrF^* + 2Ar \rightarrow ArKrF^* + Ar$	8.0×10^{-32} cm^6/sec	32,38
15	$ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$	5.0×10^{-32} cm^6/sec	33
16	$Kr + ArKrF^* \rightarrow Kr_2F^* + Ar$	2.0×10^{-11} cm^3/sec	32
17	$Ar + ArKrF^* \rightarrow Ar_2F^* + Kr$	2.0×10^{-11} cm^3/sec	32
18	$KrF^* + Kr + Ar \rightarrow Kr_2F^* + Ar$	6.5×10^{-31} cm^6/sec	13
19	$ArKrF^* + F_2 \rightarrow Ar + Kr + F + F_2$	1.0×10^{-9} cm^3/sec	33
20	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	3.0×10^{-10} cm^3/sec	33
21	$Kr_2^* + F \rightarrow KrF^* + Kr$	3.0×10^{-10} cm^3/sec	33
22	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.0×10^{-9} cm^3/sec	33
23	$Ar_2^* + F_2 \rightarrow Ar_2F^* + F$	2.5×10^{-10} cm^3/sec	33
24	$Ar_2^* + F \rightarrow ArF^* + Ar$	3.0×10^{-10} cm^3/sec	33
25	$Ar_2F^* + F_2 \rightarrow 2Ar + F + F_2$	1.0×10^{-9} cm^3/sec	33
26	$Ar_2F^* + Kr \rightarrow ArKrF^* + Ar$	1.0×10^{-10} cm^3/sec	33
27	$ArKr^* + Kr \rightarrow Kr_2^* + Ar$	1.0×10^{-10} cm^3/sec	33
28	$Kr^* + 2Ar \rightarrow ArKr^* + Ar$	1.0×10^{-32} cm^6/sec	33
29	$Ar^* + Kr + Ar \rightarrow ArKr^* + Ar$	1.0×10^{-32} cm^6/sec	33
30	$Kr^* + Kr + Ar \rightarrow Kr_2^* + Ar$	1.0×10^{-32} cm^6/sec	33
31	$Ar^* + 2Ar \rightarrow Ar_2^* + Ar$	1.14×10^{-32} cm^6/sec	33
32	$Ar^* + Kr \rightarrow Kr^* + Ar$	6.2×10^{-12} cm^3/sec	9,33
33	$Ar_2^* + Kr \rightarrow Kr^* + 2Ar$	4.0×10^{-10} cm^3/sec	32
34	$Ar^+ + 2Ar \rightarrow Ar_2^* + Ar$	2.5×10^{-31} cm^6/sec	30
35	$Ar_2^+ + Kr \rightarrow Kr^+ + 2Ar$	7.5×10^{-10} cm^3/sec	31
36	$Ar^+ + Kr + Ar \rightarrow ArKr^+ + Ar$	1.0×10^{-31} cm^6/sec	39
37	$Kr^+ + 2Ar \rightarrow ArKr^+ + Ar$	1.0×10^{-31} cm^6/sec	Estimated
38	$ArKr^+ + F^- \rightarrow ArKrF^*$	1.0×10^{-6} [b] cm^3/sec	Estimated
39	$ArKr^+ + Kr \rightarrow Kr_2^+ + Ar$	3.2×10^{-10} cm^3/sec	Estimated
40	$ArKr^+ + e \rightarrow Kr^{**} + Ar$	1.0×10^{-7} cm^3/sec	Estimated
41	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.5×10^{-31} cm^6/sec	30
42	$Ar^+ + Kr \rightarrow Kr^+ + Ar$	3.0×10^{-11} cm^3/sec	31
43	$Kr^+ + Kr + Ar \rightarrow Kr_2^+ + Ar$	2.5×10^{-31} cm^6/sec	32,40
44	$F + F + M \rightarrow F_2 + M$	1.0×10^{-33} cm^6/sec	Estimated

Electron Reactions

No.	Reaction	Typical rate constant [c]	Reference
45	$F_2 + e \rightarrow F^- + F$	1.1×10^{-9} cm^3/sec	32,41
46	$Kr + e \rightarrow Kr^* + e$	4.1×10^{-11} cm^3/sec	26,22,28
47	$Kr^* + e \rightarrow Kr^{**} + e$	6.4×10^{-7} cm^3/sec	18
48	$Kr + e \rightarrow Kr^+ + 2e$	1.5×10^{-15} cm^3/sec	17,29
49	$Kr^* + e \rightarrow Kr^+ + 2e$	4.8×10^{-8} cm^3/sec	37
50	$Kr^{**} + e \rightarrow Kr^* + e$	8.0×10^{-7} cm^3/sec	18
51	$Kr^* + e \rightarrow Kr + e$	8.0×10^{-11} cm^3/sec	26,22,28
52	$Kr^{**} + e \rightarrow Kr^+ + 2e$	1.8×10^{-7} cm^3/sec	37
53	$Ar + e \rightarrow Ar^* + e$	2.8×10^{-12} cm^3/sec	26,22,28
54	$Ar^* + e \rightarrow Ar^{**} + e$	6.6×10^{-7} cm^3/sec	18
55	$Ar + e \rightarrow Ar^+ + 2e$	1.4×10^{-19} cm^3/sec	17,29
56	$Ar^* + e \rightarrow Ar^+ + 2e$	2.8×10^{-8} cm^3/sec	37
57	$Ar^{**} + e \rightarrow Ar^* + e$	9.0×10^{-7} cm^3/sec	18
58	$Ar^* + e \rightarrow Ar + e$	5.4×10^{-12} cm^3/sec	26,22,28
59	$Ar^{**} + e \rightarrow Ar^+ + 2e$	1.9×10^{-7} cm^3/sec	37
60	$Kr_2^+ + e \rightarrow Kr^{**} + Kr$	1.1×10^{-7} cm^3/sec	34,45,46
61	$Ar_2^+ + e \rightarrow Ar^{**} + Ar$	7.7×10^{-8} cm^3/sec	34,45,46
62	$F_2 + e \rightarrow 2F + e$	3.0×10^{-10} cm^3/sec	Estimated
63	$F + e \rightarrow F^-$	1.0×10^{-12} [b] cm^3/sec	Estimated

Radiation

No.	Reaction	Rate	Reference
64	$KrF^* \rightarrow Kr + F + h\nu$	1.5×10^{8}/sec	7,33
65	$ArF^* \rightarrow Ar + F + h\nu$	2.5×10^{8}/sec	7
66	$Kr_2^* \rightarrow 2Kr + h\nu$	3.3×10^{6}/sec	33
67	$Ar_2^* \rightarrow 2Ar + h\nu$	3.8×10^{6}/sec	33
68	$Kr_2F^* \rightarrow 2Kr + F + h\nu$	6.7×10^{7}/sec	33
69	$Ar_2F^* \rightarrow 2Ar + F + h\nu$	2.0×10^{8}/sec	33
70	$ArKr^* \rightarrow Ar + Kr + h\nu$	3.0×10^{6}/sec	33
71	$ArKrF^* \rightarrow Ar + Kr + F + h\nu$	5.0×10^{7}/sec	33

Absorption

No.	Reaction	Cross section	Reference
72	$F_2 + h\nu \rightarrow F + F$	1.2×10^{-20} cm^2	42
73	$F^- + h\nu \rightarrow F + e$	5.0×10^{-18} cm^2	47
74	$Kr_2^+ + h\nu \rightarrow Kr^+ + Kr$	1.5×10^{-17} cm^2	24
75	$Ar_2^+ + h\nu + Ar^+ + Ar$	1.0×10^{-17} cm^2	24
76	$Ar^{**} + h\nu \rightarrow Ar^+ + e$	2.3×10^{-18} cm^2	48
77	$Kr^{**} + h\nu \rightarrow Kr^+ + e$	4.5×10^{-18} cm^2	48
78	$Ar^* + h\nu \rightarrow Ar^+ + e$	1.0×10^{-19} cm^2	48
79	$Kr^* + h\nu \rightarrow Kr^+ + e$	3.2×10^{-20} cm^2	48
80	$Kr_2F^* + h\nu \rightarrow$ products	5.0×10^{-18} cm^2	Estimated
81	$Ar_2F^* + h\nu \rightarrow$ products	1.0×10^{-18} cm^2	Estimated
82	$KrF^* + h\nu \rightarrow Kr + F + 2h\nu$	2.4×10^{-16} cm^2	70

[a] Definitions for tables: Intrinsic efficiency—optical energy out divided by energy deposited in the gas. g_0/g_0—ratio of small-signal gain (cm^{-1}) to total absorption coefficient (cm^{-1}). Discharge enhancement ratio—energy deposited by discharge divided by energy deposited by e-beam. Enhancement ratios should exceed about 3 to be profitable overall. Ionization rate ratio—ions created per unit time by the discharge divided by ions created per unit time by the e-beam. When this ratio exceeds about 0.5, the discharge begins to enter the ionization-dominated regime; when it exceeds unity, the discharge is self-sustained. Ion channel pumping—percentage of KrF* molecules formed through reaction 4 and reaction 5 (followed by reaction 2).

[b] Expressed as effective two-body rate constant for 2 atm pressure. Pressure dependence available graphically in reference.

[c] Using the conditions of Case 1, Table III. The cross sections for the electron-impact processes vary considerably with electron energy. Therefore the associated rate constants depend strongly on the ratio of the electric field strength to gas number density and to a lesser extent on electron density.

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(2) $ArF^* + Kr \rightarrow KrF^* + Ar$ (displacement),

(3) $Kr^+ + F^- + M \rightarrow KrF^* + M$ (neutralization).

For gases with relatively low concentrations of krypton direct electron-beam-pumping falls into two distinct regimes that are ultimately related to be the dominant electron loss process. The dominant electron loss process is, in turn, related to the electron-beam current density. Hard beam pumping (say, thousands of A/cm^2) results in large electron and ion densities and electron-ion recombination dominates the electron loss. Moderate beam pumping (say, 50 A/cm^2 or less) results in much lower electron densities so dissociative electron attachment to F_2 dominates the electron loss. The large Kr* and Ar* densities produced by electron-ion recombination at high electron-beam current densities cause the neutral chemistry of reactions 1, 2, 4, 6, and 7 to dominate KrF* production. On the other hand, the ion chemistry of reactions 3 and 5 dominates the KrF* production at low electron-beam current densities.

Finally, one would hope to gain some energetic advantage by pumping in a discharge mode, thus putting energy directly into the argon and krypton metastable species and not wasting the additional energy needed to form ions which must be neutralized later anyway. (15.7 eV to produce Ar^+ versus 11.6 eV to produce Ar*. Only 5 eV of this energy will emerge from KrF* as a photon.) Thus, one would expect the neutral chemistry to dominate a properly designed discharge, and KrF* to be formed chiefly through reactions 1, 2, and 4. In actual midrange operation of e-beam-sustained discharges (low beam current, moderate discharge voltage), simulation results demonstrate that about 30% of the energy enters the lasing state through ionic channels.

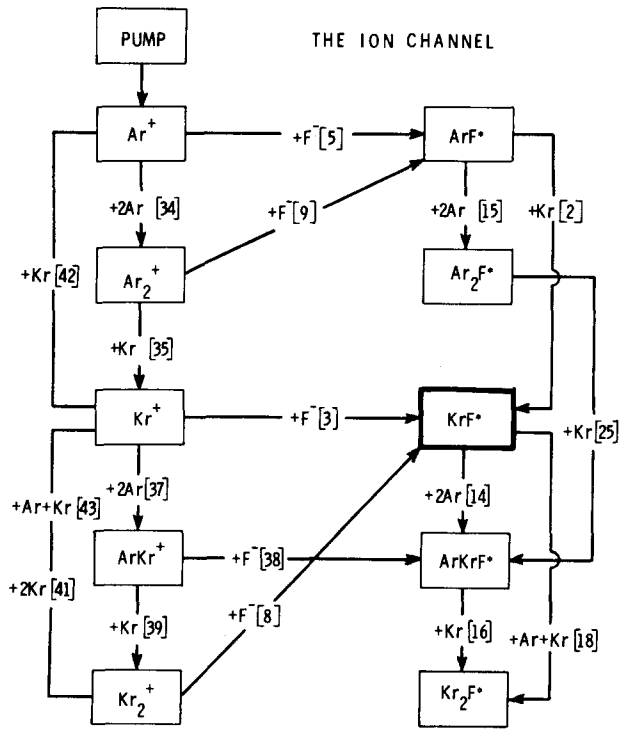


FIG. 1. Formation of krypton fluoride through the ion channel.

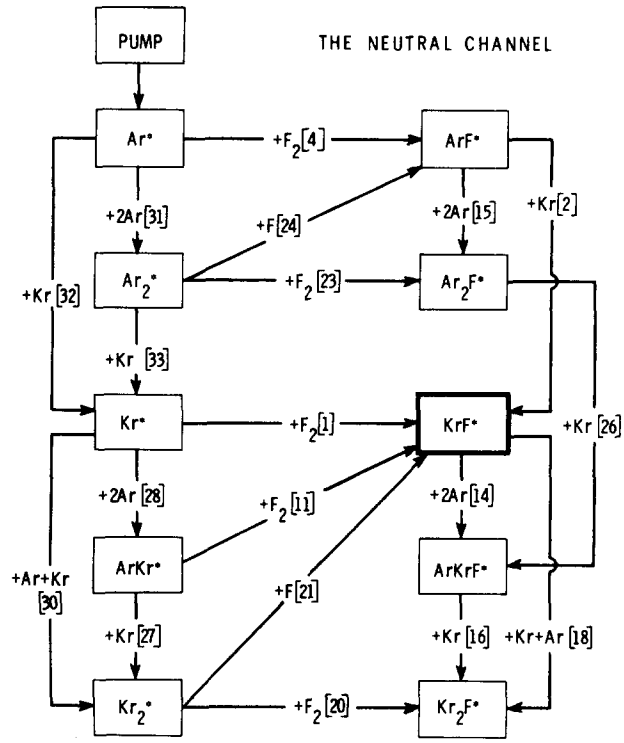


FIG. 2. Formation of krypton fluoride through the neutral channel.

A good conceptual picture of the similarities and differences of the ion and netural "channels" for pumping the KrF* molecule is presented in Figs. 1 and 2; this schematic form is due to Huestis and Lorents at Stanford Research Institute.[62] In Figs. 1 and 2 reactions (on both sides) moving down the page involve the addition of rare gas molecules; reactions moving across the page involve addition of fluorine. The appropriate reaction numbers appear in brackets next to the additives.

One can clearly see in Figs. 1 and 2 the question which dominated early development work on KrF lasers: What processes control the net amount of KrF available for stimulated emission in whatever channel is appropriate? If the controlling mechanism is quenching of an already formed KrF* molecule—that is, reactions 14 and 18, leading to Kr_2F^* (plus fluorine quenching, reaction 13)—then at least the loss rate is saturable (it depends directly on KrF* concentration which can be kept low with a large radiation field). But, if the controlling mechanism is "interception"—that is, reactions 15, 28, and 30 in the neutral channel and reactions 15, 37, 41, and 43 in the ion channel—then pumped energy is diverted into other species before KrF* is ever formed; this loss of energy is not saturable and would thus present a severe limitation to the efficiency of a KrF* laser system.

Experiments at Maxwell Laboratories[20] and AVCO Everett Research Laboratories[21] have demonstrated that, for discharge pumping at moderate pressures ($\leqslant$2.5 atm) and for beam pumping with beams $\sim$20 A/cm^2, quenching dominates over interception. This encouraging result has been duplicated by the authors in the computational model described in Sec. IV by accurate simulations of the above

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results and by extensive parametric studies investigating the limitations of experimental unknowns in rate constants. As the system is pumped harder (by raising the pressure and discharge voltage to achieve an effectively self-sustained discharge), it tends to be driven toward interception; the higher electron densities favor the ionic channel and, in particular, increase ionic interception, forming Kr_2^+. In fact, over a great variety of experimental conditions, the numerical simulations show the Kr_2^+ density tracking upward steadily with rises in electron density, thus the intrinsic efficiency is being lowered through the reinforcing of nonsaturable loss mechanisms. (Some of this energy is recovered to the laser through reaction 8, although fluorine depletion, and thus hard pumping, tend to bound this saving. The important point is that the loss, not depending on the KrF* density, is nonsaturable.) The increase of Kr_2^+ also has serious implications in controlling absorption, as will be discussed in the next topic.

These qualitative notions are given in a quantitative basis in Sec. IV. The qualitative performance of KrF* lasers is controlled by the balance of those kinetic processes, and by several other kinetic and absorptive factors as well. The dominant ionization process in a discharge is multistep: ionization of Ar* and or Ar**; the former of these is deleterious, not merely because of its implications for discharge stability, but also because it robs the electron gas of electrons in the range (tail of the distribution function) needed to produce further Ar*. Thus, even if the ion forms KrF* by neutralization followed by displacement, the net loss to the discharge is still about 5 eV. So the discharge limitation on rare gas ion densities is also concerned with efficiency.

In an e-beam-sustained discharge mode, the total gas pressure is limited (to about 2.5 atm or so) by the fact that the dominant quenching mechanism, reaction 14, is three-body. In a strongly beam-pumped or self-sustained discharge mode, the dissociative recombination of the molecular ions become controlling reactions (reactions 60 and 61). These reactions are sensitively dependent upon temperature and also upon the states of vibrational excitation of the ions. The distribution of these vibrational states, and its variation with pumping strengths and mixtures, is not known at present, and is one of the most significant remaining uncertainties in KrF laser kinetics.

Fluorine concentration is generally limited on the lower bound (at around 0.1%) by discharge current runaway (before the production of ArF* or KrF* is slowed for lack of fluorine). Its upper bound is prescribed by discharge efficiency (ideally, F_2 should attach only enough electrons to control discharge current) and absorption of laser light by F_2 and F^-.

Some of the cross sections for electron temperature dependent reactions remain only approximately known. In most cases of interest, however, rates for these processes are known under a variety of conditions. Thus, the procedure has frequently been[22] to normalize one's Boltzmann code to the rates in known circumstances, adjusting the cross section to fit. This procedure has been followed by the present authors also, particularly in modeling reactions 45, 46, 53, 60 and 61.

B. Absorption

Photodissociation of F_2 and photodetachment of F^- (reactions 72 and 73) are present as background absorptions in all KrF laser design, though the latter can be adjusted slightly. Photoionization of rare gas metastables (reactions 76–79) is generally of minor importance, though in high-voltage discharges reactions 76 and 77 can be important enough to change the intrinsic efficiency by 0.5% or so. The remaining reactions—photodissociation (presumably) of molecular ions (reactions 74 and 75) and of triatomics (reactions 80 and 81)—are dominant for laser design. The absorption cross sections of the molecular ions have been calculated and measured,[23–25]; the cross sections for the triatomics are estimated, based on the presumption that they are similar to those of the molecular ions (being ionically bound), but with broader band and slightly lower peak absorption.

The competition between these latter two mechinsms is of considerable interest. Since the primary means of forming triatomics is through quenching of KrF* and ArF*, their absorption is mostly saturable; processes which drive up molecular ion densities cannot be saturated efficiency. Obviously, high electron density modes of operation (strong beam and ionization-dominated or self-sustained discharge) settle the matter clearly in favor of the molecular ions, and thus must overcome a serious debility. However, absorption in normal moderate discharge operation is also largely dominated by molecular ions by virtue of the high cross section. The cross sections for reactions 74 and 75 are critically functions of the state of vibrational excitation. The values shown in Table I presume an average excitation of about $v = 3$; if the molecular ions were in the ground vibrational state, this absorption would be about an order of magnitude smaller. This is a second and more pressing reason why the vibrational state of the molecular ions remains the most important unknown in these systems. The choice of high vibrational excitation made here is done for three reasons: (1) it agrees with the experiments of Hunter and McCusker[25]; (2) the vibrational deexcitation time for molecular ions (presuming that it proceeds by the mechanism of replacement of a "hot" atom in the ion by a cold one from the background gas) is long compared to their average lifetimes (terminated either by dissociative recombination or neutralization reactions); (3) this choice produces highly accurate computational simulations (Sec. IV) over a very broad range of parameters; the choice of $v = 0$ produces good agreement nowhere.

III. DISCHARGE ANALYSIS

Various aspects of the KrF laser plasma have been investigated. The earliest works related to electron energy deposition are those of Berger and Seltzer[51] and Peterson and Allen,[17] who theoretically analyzed the electron energy deposition in argon. Their results give the energy loss per unit length, $d\epsilon/dx$, along the trajectory of a beam electron. Thus, in the absence of electron scattering, one can correlate the volumetric electron production rate, S_{ion}, with the electron-beam current, j_{eb}, as

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$$S_{\text{ion}} = \frac{j_{\text{eb}}}{q_e}\left(-\frac{d\epsilon}{dx}\right)\Big/\bar{I} = \frac{j_{\text{eb}} n_{\text{gas}} \sigma_{\text{ion}}}{q_e}.$$

In this equation q_e is the electron charge, $\bar{I}$ is the energy required to produce an electron-ion pair, n_{gas} is the gas number density, and σ_{ion} is the effective ionization cross section. It is well known that the e-beam velocity distribution becomes reasonably isotropic as a result of multiple elastic scattering after the beam penetrates approximately one-fifth of a mean free path into the dense gas.[67] In order to account for the additional energy deposited in a given volume by backscattered electrons, Hunter[52] performed a series of experiments correlating e-beam energy deposition with j_{eb}. He found that it is necessary to increase the cross sections obtained from Berger-Seltzer data by a factor of between 2 and 3, depending on the electron kinetic energy. A factor of 2.4 is incorporated in the present simulation model.

Mangano and Jacob[18] performed the first KrF discharge experiments. In a subsequent paper, Daugherty *et al.*[19] focused attention on the runaway discharge current which occurs in KrF plasmas unless the discharge voltage is removed quickly enough. They modeled the discharge both numerically and analytically as a homogeneous plasma. The numerical simulation predicted the observed current runaway; the analytic calculation predicted that an ionization instability would occur unless the electron attachment rate is larger than twice the field-driven ionization rate.

Hunter and Johnson[53] studied the KrF discharge equilibrium. We concluded that the attachment-dominated discharge can operate in either of two equilibria that are characterized by the dominant loss mechanism of the rare gas metastable states. Both equilibria have been observed in electron-beam-pumped discharges.[52] The eventual loss of equilibrium was attributed to the depletion of F_2 through a variety of loss processes. The two equilibria were shown to have different sensitivities—one strong and one weak—to the F_2 concentration.

Long[54] pointed out the important effects of electron-electron collisions on the electron mobility and the metastable production efficiency at low E/N.

Nighan[55] has refined the plasma stability analysis for discharges with external ionization sources to include both s and p atomic states and also a variation of the ionization rate constant with changes in fractional ionization. He discusses the increase in the F_2 attachment rate constant with increasing F_2 vibrational excitation, pointing to the stabilizing influence of the increasing rate constant.

In the following analysis, we improve upon our earlier discharge analysis by including direct e-beam metastable state excitation and elaborating on the abrupt loss of equilibrium due to F_2 depletion. The purpose of the model is to *interpret* general discharge features, not to predict discharge evolution quantitatively. We demonstrate that, within the framework of linear stability theory for unbounded homogeneous plasma, the attachment-dominated KrF discharge is stable against fluctuations in electron density for fractional ionization less than 10^{-5}. Experimentally observed current runaway is attributed to inevitable loss of equilibrium rather than an unstable equilibrium that might be stabilized experimentally.

A. Model

A subset of the processes tabulated in Table I are considered explicitly in an analytic model of the electron-beam-sustained KrF discharge. Those reactions which appear to be the controlling processes for discharges with electron number densities less than approximately 5×10^{14} cm^{-3} are selected; the validity of the selection has been verified by detailed simulations. The species which determine the discharge properties are electrons, Kr*, Kr**, Ar*, Ar**, and F_2. Continuity equations for the dominant discharge species can be written using the designated processes:

Electrons

$$\dot{n}_e = \underbrace{S_1 + S_2}_{\text{e-beam ionization}} - \underbrace{R_{45} n_e n_{F_2}}_{\text{electron attachment}} + \underbrace{n_e(R_{49} n_{Kr^*} + R_{56} n_{Ar^*} + R_{52} n_{Kr^{**}} + R_{59} n_{Ar^{**}})}_{\text{field-driven ionization}}; \tag{1}$$

Kr*

$$\dot{n}_{Kr^*} = S_4 + R_{46} n_e n_{Kr} + R_{50} n_e n_{Kr^{**}} - n_{Kr^*}(R_{49} n_e + R_{47} n_e + R_1 n_{F_2}); \tag{2}$$

(e-beam excitation; field-driven excitation; superelastic collisions; ionization; cumulative excitation; excimer formation)

Ar*

$$\dot{n}_{Ar^*} = S_3 + R_{53} n_e n_{Ar} + R_{57} n_e n_{Ar^{**}} - n_{Ar^*}(R_{56} n_e + R_{54} n_e + R_4 n_{F_2}); \tag{3}$$

Kr**

$$\dot{n}_{Kr^{**}} = R_{47} n_e n_{Kr^*} - n_{Kr^{**}}[n_e(R_{52} + R_{50}) + R_6 n_{F_2}]; \tag{4}$$

(field-driven excitation; ionization; superelastic collisions; excimer formation)

Ar**

$$\dot{n}_{Ar^{**}} = R_{54} n_e n_{Ar^*} - n_{Ar^*}[n_e(R_{59} + R_{57}) + R_7 n_{F_2}], \tag{5}$$

where R_{xx} is the kinetic rate constant for process xx,

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$$S_1 = \frac{j_{eb}\sigma_1 n_{Ar}}{q_e}, \quad S_2 = \frac{j_{eb}\sigma_2 n_{Kr}}{q_e},$$
$$S_3 = \frac{j_{eb}\sigma_3 n_{Ar}}{q_e}, \quad S_4 = \frac{j_{eb}\sigma_4 n_{Kr}}{q_e}.$$

The justification of the method of including direct electron-beam excitation is based on the analysis of Holmes *et al.*[56] The cross sections σ_1, σ_2, σ_3, and σ_4 were obtained by increasing the Berger-Seltzer ionization cross sections by a factor of 2.4 [52] and assuming an excitation partitioning of 75% into ionization and 25% into excitation.[17,66] Krypton and argon were considered to be electronically indistinguishable with regard to partitioning. The cross sections are $\sigma_1 = 7.68\times10^{-18}$ cm^2, $\sigma_2 = 1.92\times10^{-17}$ cm^2, $\sigma_3 = 2.64\times10^{-18}$ cm^2, and $\sigma_4 = 4.22\times10^{-18}$ cm^2. These equations can be solved to determine both the conditions that permit steady-state solutions to exist and the approximate temporal evolution of equilibrium. The stability of the equilibria can be studied once conditions for their existence are determined.

B. Equilibria

The electron number density is determined by the balance between neutral species ionization and electron attachment. Since F_2 is continously depleted, the electron attachment rate diminishes as time progress. The electron number density increases as the electron loss rate decreases. Thus, steady-state operation is never quite achieved.

The F_2 depletion rate is generally slower than the rates of ion processes by at least an order of magnitude. Two characteristic time scales govern the equilibrium and stability of the discharge. The ground-state neutral species vary with times on the order of microseconds, while the electronically excited and ionic species vary with times on the order of tens of nanoseconds. As a result, equilibrium can be achieved approximately for relatively short times, even though long-term equilibrium for the attachment-dominated discharge is not possible. Subsequent discussions of equilibrium refer to temporal evolution of metastable and ionic species.

The rate equations for the metastable species indicate that two forms of equilibrium can exist. These equilibria correspond to two limiting cases of metastable species loss processes: (a) loss dominated by excimer formation and (b) loss dominated by metastable ionization. Both equilibria will be examined with respect to discharge stability.

When the metastable species losses are dominated by excimer formation, the rate equations yield approximately

$$n_{Kr^*} \approx \frac{S_4 + R_{46} n_e n_{Kr}}{2R_1 n_{F_2}}, \quad n_{Ar^*} \approx \frac{S_3 + R_{53} n_e n_{Ar}}{2R_4 n_{F_2}},$$
$$n_{Kr^{**}} \approx \frac{R_{47} n_e n_{Kr^*}}{2R_6 n_{F_2}}, \quad n_{Ar^{**}} \approx \frac{R_{54} n_e n_{Ar^*}}{2R_7 n_{F_2}}.$$

KrF discharges are often operated near the upper limit of the excimer formation equilibrium where the excimer formation and ionization losses are nearly equal. The factors of $\frac{1}{2}$ are included in the equations above to estimate the loss rate, and hence the excited-state densities, more precisely in this region. The equilibrium electron number density evolves from the modified electron rate equation:

$$\begin{aligned} n_e = {} & S_1 + S_2 + n_e\left(\frac{R_{49}S_4}{2R_1 n_{F_2}} + \frac{R_{56}S_3}{2R_4 n_{F_2}} - R_{45} n_{F_2}\right) \\ & + n_e^2\left(\frac{R_{46}R_{49}n_{Kr}}{2R_1 n_{F_2}} + \frac{R_{53}R_{56}n_{Ar}}{2R_4 n_{F_2}}\right. \\ & \left. + \frac{R_{47}R_{52}n_{Ar}}{4R_1R_6 n_{F_2}^2} + \frac{R_{54}R_{59}S_3}{4R_4R_7 n_{F_2}^2}\right) \\ & + n_e^3\left(\frac{R_{46}R_{47}R_{52}n_{Kr}}{4R_1R_6 n_{F_2}^2} + \frac{R_{53}R_{54}R_{59}n_{Ar}}{4R_4R_7 n_{F_2}^2}\right). \end{aligned} \tag{6}$$

The equilibrium solution for n_e can be obtained from Eq. (6) with $\dot{n}_e = 0$, and the time-dependent equation can be integrated directly. Unfortunately, the resulting expressions are not trivial.

The purpose of analytic modeling is to illustrate the physical character of the discharge processes; therefore, any cumbersome solutions must be rejected since they obscure rather than clarify the problem under consideration. The simplification that follows is introduced in the spirit of clarifying the discharge analysis. No important physical details are sacrificed since the approximations are chosen to given the least errors under typical discharge operating conditions.

The maximum electron density for which Eq. (6) is valid (because excited-state losses are dominated by ionization instead of excimer formation at larger electron densities) is

$$n_e^{max} \approx \frac{R_1 n_{F_2}}{R_{47}} \quad \text{or} \quad n_e^{max} \approx \frac{R_4 n_{F_2}}{R_{54}}.$$

Using these values since they occur often in experiments, the terms $R_{52}n_{Kr^{**}}$ and $R_{59}n_{Ar^{**}}$ in Eq. (1) are approximately $R_{52}n_{Kr^{**}} \approx R_{52}n_{Kr^*}/2$ and $R_{59}n_{Ar^{**}} \approx R_{59}n_{Ar^*}/2$, respectively. It follows that the electron rate equation is

$$\dot{n}_e \approx S_1 + S_2 - \alpha n_e + \beta n_e^2, \tag{7}$$

where

$$\alpha \equiv R_{45}n_{F_2} - \left(\frac{S_4(2R_{49}+R_{52})}{4R_1 n_{F_2}} + \frac{S_3(2R_{56}+R_{59})}{4R_4 n_{F_2}}\right) \tag{8}$$

and

$$\beta \equiv \frac{R_{46}(2R_{49}+R_{52})n_{Kr}}{4R_1 n_{F_2}} + \frac{R_{53}(2R_{56}+R_{59})n_{Ar}}{4R_4 n_{F_2}}. \tag{9}$$

The modified electron rate equation determines the conditions for discharge equilibrium. In steady state,

$$n_e = \frac{\alpha}{2\beta}\left[1 - \left(1 - \frac{4\beta(S_1+S_2)}{\alpha^2}\right)^{1/2}\right]. \tag{10}$$

This electron density is achieved in the *excimer formation equilibrium*, where the metastable species losses are dominated by excimer formation. For equilibrium to exist, it is necessary that

$$\alpha^2 \geq 4\beta(S_1+S_2). \tag{11}$$

The excimer formation equilibrium is a condition corresponding to an electron-beam-dominated discharge, as will now be shown. The maximum electron density consistent with equilibrium is achieved when $\alpha^2 = 4\beta(S_1+S_2)$

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and Eq. (10) yields

$$n_e^{\max} = \alpha/2\beta ; \tag{12}$$

therefore, $\beta n_e^2 \leqslant S_1 + S_2$. Hence, the field-driven electron source term, βn_e^2, is at most equal to the electron source term due to the electron beam, $S_1 + S_2$.

The equilibrium condition, Eq. (11), establishes a relationship between the maximum values of applied electric field strengths and electron-beam currents. The coefficient α is the electron attachment rate less the two-step ionization rate of Ar and Kr driven by electron-beam excitation of a metastable state followed by field-driven ionization. The coefficient β represents the rate constant for two-step field-driven ionization, roughly proportional to $(E/N)^2$. The source terms, S_1 and S_2, represent the volumetric rate of direct electron-beam ionization. As a result, α decreases linearly with j_{eb} and $S_1 + S_2$ increases linearly with j_{eb}. Hence, for each E/N there is a maximum j_{eb} for which the excimer formation equilibrium can exist. Conversely, for a fixed electron-beam current, there is a maximum value of E/N at which the discharge can operate without making a transition to some other regime where the ionization is produced primarily by the electric field.

The electron rate equation (7), where the metastable Ar and Kr losses are dominated by excimer formation, can be integrated to yield

$$n_e(t) = 2(S_1 + S_2)\left(\alpha\left\{1 + \left(1 - \frac{4\beta(S_1 + S_2)}{\alpha^2}\right)^{1/2} \times \coth\left[\frac{\alpha t}{2}\left(1 - \frac{4\beta(S_1 + S_2)}{\alpha^2}\right)^{1/2}\right]\right\}\right)^{-1} \tag{13}$$

for $\alpha^2 \geqslant 4\beta(S_1 + S_2)$, and

$$n_e(t) = 2(S_1 + S_2)\left(\alpha\left\{1 + \left(\frac{4\beta(S_1 + S_2)}{\alpha^2} - 1\right)^{1/2} \times \cot\left[\frac{\alpha t}{2}\left(\frac{4\beta(S_1 + S_2)}{\alpha^2} - 1\right)^{1/2}\right]\right\}\right)^{-1} \tag{14}$$

for $\alpha^2 < 4\beta(S_1 + S_2)$. The excimer formation equilibrium emerges from Eq. (13) with a characteristic time

$$\tau_{exc} = \frac{2}{[\alpha^2 - 4\beta(S_1 + S_2)]^{1/2}} . \tag{15}$$

Under the field-dominated conditions of Eq. (14), the electron number density is a monotonically increasing function of time. An excimer formation equilibrium is not achieved, but a second equilibrium of the attachment-dominated discharge is accessible.

Two final points regarding the excimer formation equilibrium should be noted. Even though the equilibrium condition $\alpha^2 \geqslant 4\beta(S_1 + S_2)$ can be satisfied immediately upon discharge initiation, continuous F_2 depletion results in a decreasing α and an increasing $\alpha^2/\beta \sim n_{F_2}^3$ so that eventually the discharge must make a transition to some other type of equilibrium condition. The transition may also result from a plasma ionization instability, as is discussed in Sec. III C.

If the electron density becomes so large that the metastable Ar and Kr loss rates are dominated by ionization instead of excimer formation, Eqs. (1)–(5) yield approximately

$$n_{Kr^{**}} \approx \frac{R_{47} n_{Kr^*}}{2(R_{52} + R_{50})}, \qquad n_{Ar^{**}} \approx \frac{R_{54} n_{Ar^*}}{2(R_{59} + R_{57})},$$

$$n_e n_{Kr^*} \approx \frac{2(R_{52} + R_{50})(S_4 + R_{46} n_e n_{Kr})}{4(R_{49} + R_{47})(R_{52} + R_{50}) - (R_{50} R_{47})},$$

and

$$n_e n_{Ar^*} \approx \frac{2(R_{59} + R_{57})(S_3 + R_{53} n_e n_{Ar})}{4(R_{56} + R_{54})(R_{59} + R_{57}) - (R_{57} R_{54})} .$$

Using these approximations for the metastable state number densities, one can reduce the electron rate equation to

$$\dot{n}_e \approx S_0 - \gamma n_e , \tag{16}$$

where

$$S_0 \equiv S_1 + S_2 + C_1 S_3 + C_2 S_4 , \tag{17}$$

$$\gamma \equiv R_{45} n_{F_2} - C_1 R_{53} n_{Ar} - C_2 R_{46} n_{Kr} , \tag{18}$$

$$C_1 = \frac{2R_{56}(R_{59} + R_{57}) + R_{54} R_{59}}{4(R_{56} + R_{54})(R_{59} + R_{57}) - R_{57} R_{54}} , \tag{19}$$

and

$$C_2 = \frac{2R_{49}(R_{52} + R_{50}) + R_{47} R_{52}}{4(R_{49} + R_{47})(R_{52} + R_{50}) - R_{50} R_{47}} . \tag{20}$$

The term γ is the electron attachment rate minus the cumulative ionization rate. In steady state,

$$n_e = S_0/\gamma \tag{21}$$

for the *ionization equilibrium*. The equilibrium can be achieved if $\alpha^2 < 4\beta(S_1 + S_2)$ and if $\gamma > 0$.

In contrast with the excimer formation equilibrium, the ionization equilibrium corresponds to a field-dominated discharge. More electrons are produced by cumulative field-driven ionization than directly by the electron beam. In addition, the equilibrium condition, $\gamma > 0$, is not strongly constrained by the magnitude of the electron-beam current; the condition simply requires that the electron attachment rate exceed the cumulative ionization rate. Since γ diminishes in time as a result of continuous F_2 depletion, it would seem reasonable to decrease n_e primarily by decreasing j_{eb} (and hence S_0), rather than by "fine tuning" the E/N to increase γ. Note, however, that the ionization equilibrium condition, which is proportional to n_{F_2}, is geneally less sensitive to changes in F_2 density than the excimer formation equilibrium condition which is proportional to $n_{F_2}^3$. In addition, the ratios of metastable state densities (e.g., $n_{Kr^{**}}/n_{Kr^*}$) are essentially independent of n_e.

The electron rate equation (16) can be integrated to yield for the ionization equilibrium

$$n_e(t) = (S_0/\gamma)[1 - \exp(-\gamma t)] . \tag{22}$$

The equilibrium is established in a characteristic time

$$\tau_{ion} = \gamma^{-1} . \tag{23}$$

The complete temporal evolution of the attachment-dominated discharge can now be examined. Either the excimer formation equilibrium or the ionization equilibrium can be achieved within characteristic times τ after discharge initiation. If the excimer formation equilibrium is obtained, the electron and metastable species will cause a gradual depletion of the F_2 density according to

$$\dot{n}_{F_2} = -\tau_{F_2}^{-1} n_{F_2} , \tag{24}$$

where $\tau_{F_2}^{-1} \approx R_{45} n_e + R_1 n_{Kr^*} + R_4 n_{Ar^*}$. As a result, α will

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decrease, β will increase, and the discharge will make a transition to the ionization equilibrium. The depletion of F_2 then proceeds according to

$$\dot{n}_{F_2} \approx -\frac{S_0 n_{F_2}}{n_{F_2} - n_u} - Q_e n_{F_2} \tag{25}$$

for $n_{F_2} \gg n_u$, where n_u is the value of n_{F_2} for which $\gamma = 0$ and $Q_e = R_1 n_{Kr^*} + R_4 n_{Ar^*}$ is the F_2 quenching rate due to excimer formation. As we see in Eq. (25), F_2 is depleted gradually, approximately as

$$\dot{n}_{F_2} \approx -[S_0 + Q_e n_{F_2}(t=0)] \tag{26}$$

until n_{F_2} approaches n_u. At that time the F_2 loss rate increases dramatically due to attachment by the rapidly increasing electron density ($\gamma \to 0$) the attachment-dominated equilibrium ceases to exist at approximately

$$t_{eq} = \frac{n_u}{S_0}\left[\ln\left(\frac{n_u}{n_0}\right) + \frac{n_0}{n_u} - 1\right], \tag{27}$$

where $n_0 = n_{F_2}(t=0)$. The cumulative ionization rate exceeds the electron attachment rate for $n_{F_2} < n_u$, and the electron density grows until it is stopped by two-body or three-body recombination processes.

The general features of equilibrium behavior are illustrated in Fig. 3, which was generated by computational simulation of the KrF discharge and has the features predicted by the analytic model. Electron density rises quickly to the upper limit of the excimer equilibrium and then grows slowly for a few hundred nanoseconds. Eventually, as the F_2 is depleted, the discharge enters the ionization equilibrium.

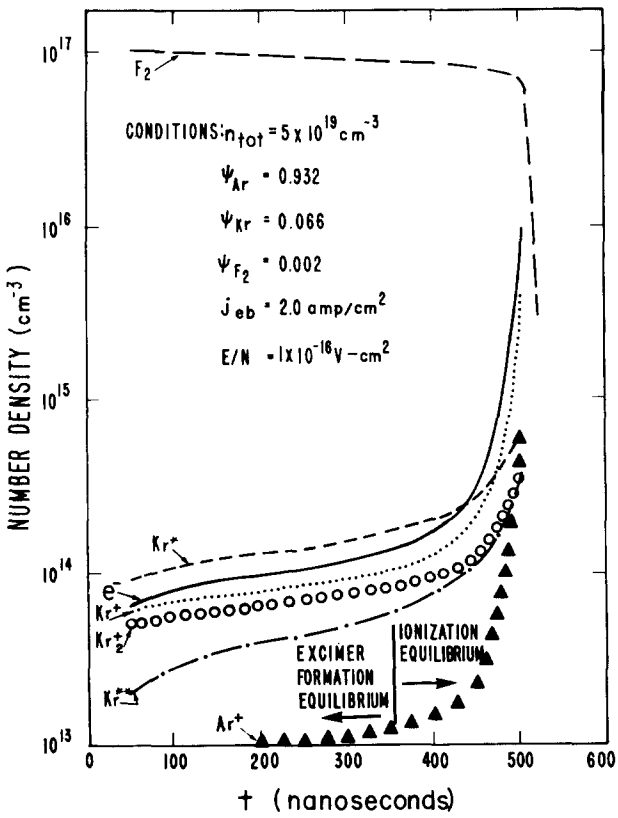


FIG. 3. Temporal evolution of species densities. Current runaway occurs at 530 nsec.

The F_2 depletion can be relatively minor since the excimer equilibrium limit is proportional to $n_{F_2}^3$. At first the electron density increases rather slowly, reflecting the continuing gradual F_2 loss predicted by Eq. (26), and then increases abruptly (note the "knee" in the F_2 curve in Fig. 3) as n_{F_2} approaches n_u [the F_2 depletion rate in Eq. (25) increases dramatically] and attachment can no longer restrain the growing electron population. The electron density increases more than one order of magnitude before being stopped by recombination when small fractions of Kr are used in the gas mixture. Since Kr has a lower ionization potential than Ar, the sustainer field (which dominates the ionization at late times) produces primarily Kr^+ ions. And since $n_{Kr^+} \approx n_e$, the ratio of the electron-ion recombination rate to electron attachment rate is approximately $R_{41} n_{Kr}^2 / R_{45} n_{F_2} = 0.02$ for the mixture used in the computation. Thus the electron density would have to increase to more than $10^{16}\,cm^{-3}$ in order to equal the loss rate due to attachment at $n_e = 2 \times 10^{14}\,cm^{-3}$. However, since the metastable production rate and hence the ionization rate is proportional to n_e, the electron growth is not likely to be stopped before it reaches densities characteristic of arc discharges.

C. Stability

We have determined two distinguishable equilibria for the attachment-dominated KrF discharge. Now we will consider their stability against small density fluctuations.

A simple linear stability analysis is performed on Eqs. (7) and (16). The electron density is decomposed into equilibrium and fluctuating components, denoted as n_0 and n_1, respectively. The linearized equations are

$$\dot{n}_1 = -\alpha n_1 + 2\beta n_0 n_1 \tag{28}$$

for the excimer formation equilibrium and

$$\dot{n}_1 = -\gamma n_1 \tag{29}$$

for ionization equilibrium. The growth or decay of a perturbation is determined by assuming that n_1 is proportional to $\exp(\nu t)$ and solving Eqs. (28) and (29) for ν. For the excimer formation equilibrium,

$$\nu_{exc} = 2\nu_i - \alpha, \tag{30}$$

where $\nu_i = \beta n_0$ is the field-driven ionization rate, and for the ionization equilibrium

$$\nu_{ion} = -\gamma. \tag{31}$$

An equilibrium is unstable for positive ν and stable for negative ν.

We see that the excimer formation equilibrium is stable for

$$\alpha > 2\nu_i. \tag{32}$$

In the limit of small E/N, α becomes the attachment rate per electron, ν_a, and the inequality (32) reduces to the stability criterion derived by Daugherty *et al.*[19]

It is also important to note from Eqs. (11) and (12) that at the upper limit for the existence of the excimer formation equilibrium $n_0 = n_e^{max}$, $\beta n_0 = \frac{1}{2}\alpha$, or

$$\alpha = 2\nu_i. \tag{33}$$

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TABLE II. *e*-beam direct-pumped laser experiments. Experiments performed at Maxwell Laboratories, Inc.,[20] AVCO-Everett Research Laboratories,[21] and Northrop Research and Technology Center.[51]

Experimental parameters	AVCO	Maxwell	Northrop
Total pressure (atm)	1.7	1	1.83
Mix—Ar/Kr/F_2	95.8/4/0.2	93.3/6.6/0.1	92.7/7.2/0.1
Beam average current density (Amps/cm^2)	11.5	6	15-18
Beam voltage (kV)	250	300	300
Electrical pulse length (nsec)	600	500	900
Volume (liters)	8.5	60	3

	AVCO		Maxwell		Northrop	
Results	Expt	Simulation [a]	Expt	Simulation	Expt	Simulation [b]
Optical output (J)	102	104.6	120	124.8	54	52.65
Average optical flux (W/cm^2)	?	2.3×10^6	7×10^5	8.1×10^5	?	7.8×10^6
Intrinsic efficiency	9.8%	9.8%	7.2%	8.0%	15%	11.25%
Total absorption	0.27%/cm	0.23%/cm	?	0.39%/cm	?	1.2%/cm
g_0/g_a	~10	12	~6	9.3	?	5.1
Ion channel pumping	?	57%	?	43%	?	60%

[a] The power input to the gas was tailored specifically in the code to match the estimated total power deposited by the beam.
[b] The lower estimate of beam current (15 A/cm^2), taken as a step function over the electrical pulse time, was used in the simulation.

When $\alpha = 2\nu_i$ the discharge transitions from the excimer formation equilibrium to the ionization equilibrium. In other words, the discharge equilibrium does not become unstable when ν_i increases to the point that $\alpha = 2\nu_i$. Instead, the equilibrium changes its character and becomes stable against fluctuations in electron density. Thus, we conclude that the KrF discharge is always stable in the attachment-dominated regime.

It is possible that the discharge may become unstable as a result of physical mechanisms that are not included in this elementary analysis. A more sophisticated stability analysis should incorporate energy deposition, spatial inhomogeneities, and power supply constraints. We note, however, that in some cases, such as Ref. 18, an equilibrium computational simulation appears to describe quite accurately the discharge current measured in an experiment that was operated in a regime labled unstable for the excimer formation equilibrium. That should not occur if the instability growth rate is of the same order of magnitude as ν_a, or about 10^8 sec^{-1}. In addition, the stable experiment in the ionization equilibrium that Hunter performed[65] further demonstrates that laboratory experiments are feasible even when $\alpha < 2\nu_i$.

IV. SIMULATIONS

A. Kinetics and power extraction

In this section we present results of a numerical simulation of KrF laser kinetics and power extraction. The purpose is to demonstrate the sufficient workability of the rate system displayed in Table I (and the associated numerical solution), to demonstrate by example the mechanisms described in II, and to demonstrate the inherent power limitation of systems as yet not constructed. In Sec. V, numerical simulations of optical properties of these lasers will be coupled to these simulations.

A complete description of the code developed for KrF appears elsewhere,[49] but it is necessary to present here at least an outline of its methods. The chemical equations numbers 1–63 are solved simultaneously using a simple modified-Newton's method integrator (a version with a fourth-order Hamming predictor-corrector also exists, but it is not significantly more accurate) with simple (few percent) flux limiters on densities and their derivatives. The time step is determined by the flux limiters and by the changes in electron temperature and density. When either of these latter changes by a fraction chosen as an input parameter (usually 5%) the temperature-dependent rates 45–63 are updated by solution of a steady-state Boltzmann equation[63] (justified because the electron energy distribution function relaxes in fractions

TABLE III. *e*-beam-sustained discharge laser experiments. Both experiments performed at Maxwell Laboratories, Inc.[20]

Experimental parameters	1	2
Total pressure (atm)	2.01	1.98
Mix—Ar/Kr/F_2	95.5/4.3/0.2	93/6.6/0.4
Beam current density (A/cm^2)	2	3.5
Beam voltage (kV)	400	400
Electrical pulse length (nsec)	500	500
Discharge average electric field (kV/cm-atm)	2	2.5
Volume (liters)	30	40

Results	Expt. 1	Simulation 1	Expt. 2	Simulation 2
Optical output (J)	36	39.6	44	52
Average optical flux (W/cm^2)	7×10^5	7.8×10^5	8×10^5	1.4×10^6
Intrinsic efficiency	3%	3.4%	1.7%	2.4%
Discharge enhancement ratio	2.4	2.3	2.1	2.0
g_0/g_a	~4	3.31	?	2.9
Ionization rate ratio	?	0.136	?	0.121
Total absorption	?	0.3%/cm	?	0.4%/cm
Ion channel pumping	?	33%	?	40%

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of a nanosecond). This solution includes a self-consistent treatment of supper-elastic collisions and of electron-electron collisions.[64] The Boltzmann solution is performed to get electron-reaction rate constants even for the zero-field direct-pumped cases. At each time step, appropriate species densities are adjusted for spontaneous radiation (reactions 64–71) and stimulated emission. The KrF* gain g and the total absorption coefficient g_a are calculated, and the radiative intensity is updated by integrating.

$$\frac{dI}{dt} = cI(g - g_a - g_{\text{threshold}}) + \text{spontaneous terms}$$

along with the chemical species ($g_{\text{threshold}}$ is the threshold gain, calculated as $-(1/2L)\ln(r_1 r_2)$ where L is the gain length and $r_1 r_2$ is the product of mirror reflectivities).

Tables II, III, and IV show a representative sampling of KrF expeiments and numerical simulation results for each. The maximum energy output achieved thus far in KrF was 350 J, from Maxwell's 60-liter tank, pumped with a 10-A/cm^2 e-beam. The average expectation for energy extraction from KrF at intrinsic efficiencies in excess of 7% is about 10 J/liter. Higher values are indeed possible, but one must pay dearly in efficiency and stability.

Spatial inhomogeneities and pinching in the e-beams, nonuniform deposition and scattering of both beam and discharge electrons, and temporal inhomogeneities all degrade the ideal performances of these lasers. Applied B fields have been shown to ameliorate these difficulties in some cases.[21] But, since the code presumes essentially ideal deposition, much of the apparent high degree of agreement with experiments shown here must be taken to be fortuitous. However, in most cases the inputs were tailored to represent as closely as possible the measured total energy deposited in the gases and to simulate precisely the temporal variation of both e-beam current and discharge voltage. And these computational results are truly representative; they are not simply the best. Such numerical simulations always involve "tuning" of many parameters to approach agreement with experimental results. The code was tuned in this fashion by choosing the unknown or uncertain parameter values to provide agreement with a single experiment—case 1 in Table III. All subsequent runs were made without any further changes in these parameters. Agreement within about 20% on energy out and 30% on intrinsic efficiency was obtained for all real experimental conditions for 15 different experiments; usually agreement was much better. In five cases (including the AVCO and Northrop cases of Table II) the simulation results preceded the experiments themselves. A further simulation of fully self-sustained operation, performed to examine the ionization discharge equilibrium, duplicated (within experimental error) an unreported experiment,[65] which was only learned of some time later.

One can easily see in both direct-pumped and beam-sustained discharge cases, stronger pumping tends to drive reactions toward the ion channel, and greater absorption also results from increased densities of molecular ions. Quenching remains the dominant energy loss mechanism from KrF in the five experiments shown, and indeed in most thus far performed. Typical electron densities in all these cases tend to be about $(1\text{–}2)\times 10^{14}$ cm^{-3} in stable operation.

TABLE IV. High-power ionization-dominated discharge (proposed device).

Parameters	
Total pressure	3 atm
Mix—Ar/Kr/F_2	89.8/10/0.2
Electric pulse length	500 nsec
Beam current density	15 A/cm^2
Discharge average electric field	2.5 kV/cm-atm
Beam voltage	330 keV
Volume	20 liters
Results	
Optical output	29.7 J/liter
Average optical flux	3.44×10^6 W/cm^2
Intrinsic efficiency	4.3%
Discharge enhancement ratio	1.2
g_0/g_a	3.9
Total absorption	1.2%/cm
Discharge quality	Self-sustained
Discharge stability	Arcs ~ 600 nsec

The principal absorber is Kr_2^+ at typically 5×10^{12} to 5×10^{13} cm^{-3}.

An extreme case of strong pumping is represented in the ionization-dominated discharge of Table IV, which is a purely theoretical result (the experiment has been proposed but not performed). Here the electron density is driven above 10^{15} cm^{-3}, with the result that the Kr_2^+ density reaches 5×10^{14} cm^{-3} and absorption limits extraction efficiency severely. Ion channel pumping reaches about 70%, showing the dominance of this chain.

Neither of the two discharge experiments shown reached the desired high value of enhancement ratio, though later experiments have exceeded 3.[21] Still, discharges have not been able sufficiently to make up their losses to momentum transfer and elsewhere, and consequently the beam-only experiments have continued to provide the most promising numbers in the region 10–12%. The simulations suggest that a discharge enhancement of at least 5 would be necessary to approach such efficiencies, but none have been achieved.

The hard-pumping case of Table IV shows that it is indeed possible to get as much as 30 J/liter from a KrF e-beam-sustained discharge, if the deviations from ideality can be overcome. But these deviations are certain to become more important in such a case than in the other, relatively gently pumped cases. The result of Table IV is the very best produced in an extensive parameter study performed with the simulation code to investigate the possibilities of hard pumping. It exists in a very tight corner of the available parameter space of operation, with time rise of discharge voltage and e-beam current carefully shaped and coordinated. Raising the discharge voltage any faster (rise time is about 70 nsec) causes current runaway before significant power is extracted; raising the voltage much slower allows the molecular ion density to catch up to the rising electron density, driving the net gain down. In this middle region, with the values of beam current and discharge voltage chosen as high as possible, high joules/liter are available, but only at about

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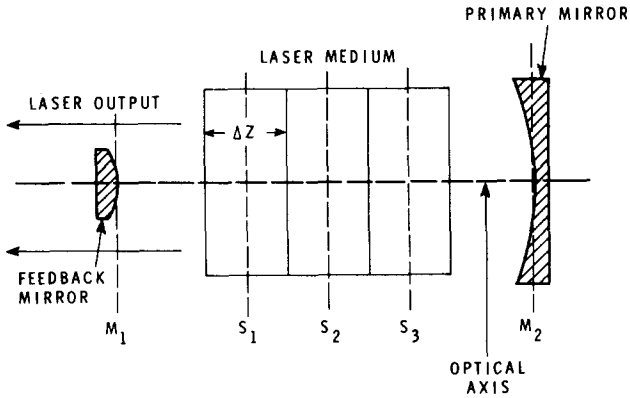


FIG. 4. Unstable optical resonator configuration. Electron beam injects from top to bottom. Electric discharge is normal to the paper.

4% efficiency. Driving the pumping mechanism so strongly toward the less efficient ion channel has taken its toll. Although absorption is very high, this can be overcome for the most part; still, the flux out is decreasing from about 400 nsec on. The laser has already begun to shut itself off by absorption when the discharge current runaway begins at about 620 nsec.

These, then, are the apparent limits of KrF performance, based on many competing mechanisms. From 10 to perhaps 15 J/liter direct pumped at 10–15% intrinsic efficiency; for discharge pumped lasers, both numbers should be lower by about 20% in a well-designed experiment.

V. PHYSICAL OPTICS

The simple intensity equation presented in Sec. IV is adequate for multimode energy extraction calculations, but it is unsuitable for investigating the optical quality of a laser. Such calculations require far more elaborate techniques to determine the intensity and phase distributions transverse to the optical axis. These calculations are performed routinely for cw infrared lasers, where the laser kinetics and optical mode evolve through interactive computations. We have adapted existing techniques for the preliminary optical investigation of the KrF laser.

A careful optical analysis of the KrF laser must be performed for several reasons. For example, some applications require lasers with short turn-on times; one might expect that the turn-on time of the laser could be reduced by operating it as a driven oscillator with injection locking, and the effects of such operation on optical quality could be calculated. It is also possible to determine the importance of possible optical disturbances caused by nonuniformity of energy deposition in direct e-beam-pumped lasers. The calculations could prescribe adaptive optical techniques required to correct for such distortions. Finally, the effects of transient concentrations of refractive species could be determined. Salvi[68] has suggested that combinations of transient absorbing and refracting species could lead to mode switching and hence to a reduction in power extraction and the ability to focus the laser.

We explore the first of these, reduction in turn-on time through injection locking, in this paper. The effects of pumping nonuniformities and transient refractive species remain to be addressed.

We have chosen to analyze an unstable optical resonator configuration because it offers the advantages of large mode volume, adjustable diffraction output coupling, and substantial discrimination against high-order transverse modes. The large mode volume is especially important for ultraviolet lasers. The fraction of energy available in the laser medium which is extracted by the lowest-order mode of a stable resonator is inversely proportional to the resonator's Fresnel number (N_F) and Fresnel numbers of the order 10^3 or 10^4 are common. An unstable resonator conversely uses the entire mode volume.

The optical resonator calculations are performed using a Fourier optics propagation algorithm developed by Salvi; details of the optics theory and the computational methods are discussed in Refs. 58–60. For computational purposes, the laser cavity is represented by a series of discrete optical elements oriented transverse to the optical axis as shown in Fig. 4. Two of these, M_1 and M_2, are the mirrors of a positive branch confocal cylindrical strip resonator. The elements, S_i, between the mirrors are gain sheets. The amplification and phase transformation that the electric field experiences in propagating through a gain region of thickness ΔZ is accounted for in a single operation at each gain sheet. The thickness ΔZ is kept sufficiently small that the average inten-

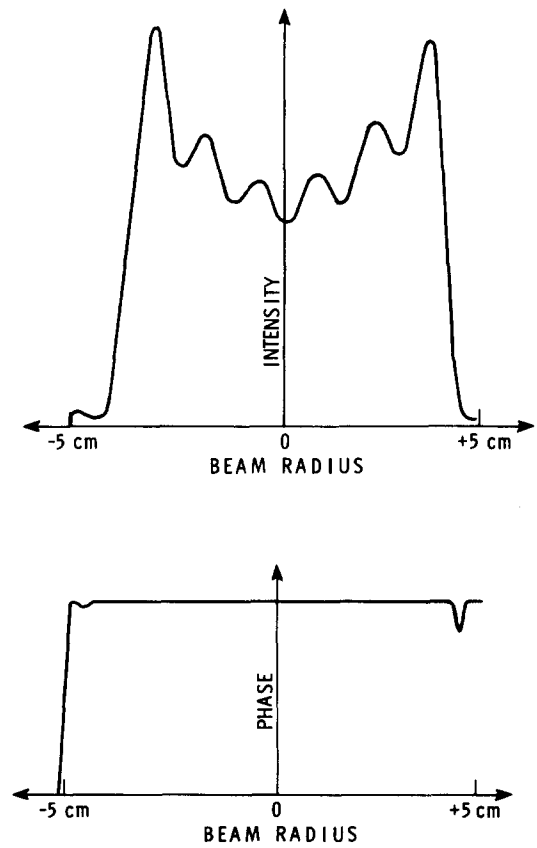


FIG. 5. Intensity and phase distributions prior to striking the feedback mirror for the direct-pumped AVCO device. Output coupling, 71%. Electron-beam current density, 11.5 A/cm^2.

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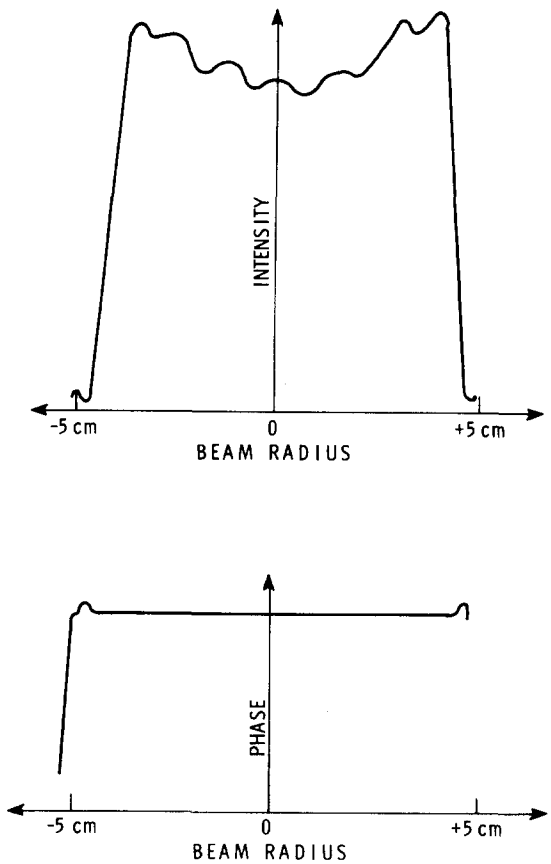


FIG. 6. Intensity and phase distributions prior to striking the feedback mirror for the electron beam sustained Maxwell device. Output coupling, 80%. Electron-beam current density, 2.0 A/cm^2. $E/N = 8\times10^{-17}$ V cm^2.

sity within a gain region is approximated well by $I_i^{\pm}[\exp(\alpha\Delta Z)-1]/\alpha\Delta Z$. The intensity entering gain region i from the left is I_i^+, and the corresponding intensity entering from the right is I_i^-. The coefficient α is $g-g_a$. We use three gain sheets in the calculations shown, the plasma being represented by 30 fluid cells coupled only by the laser flux (although the code is capable of treating one-dimensional gas dynamic flow integrally, where such flow is necessary for modeling a particular laser).

Spontaneous emission incoherence during laser initiation is accounted for by uniformly randomizing the optical phase at all computational points of each gain sheet. Injection locking is simulated by adding an initial optical component of unform plane wave to the spontaneous emission.

Injection locking with amplitudes ranging from zero up to 10% of the oscillator cavity flux was simulated. The intensity and phase distributions, with and without injection locking, were indistinguishable after a few photon round trips in the cavity. This is probably due to the large rejection of high spatial frequencies by the high-loss resonator. We conclude that injection locking does not measurably reduce the time required to establish a stable optical mode. There is evidence, however, that injection locking can be used efficiently in reducing the spectral width.[61]

Intensity and phase distributions at 200 nsec for the AVCO experiment in Table II and the Maxwell experiment 1 in Table III are shown in Figs. 5 and 6, respectively. These distributions are associated with the outbound electric field immediately prior to striking the feedback mirror. The intensity distribution is optically underresolved since a grid containing more than 30 000 computational points transverse to the optical axis is required to resolve each of the resonator Fresnel zones adequately. The larger number of points (we used a maximum of 512) serves to resolve the small-amplitude high-frequency oscillations in the electric field. These oscillations are unimportant in power extraction calculations and their resolution enormously increases computational storage requirements. For the same reason, we used cylindrical mirrors instead of spherical mirrors in studying mode build-up time. The most important feature to observe in Figs. 5 and 6 is the uniform optical phase that should occur for large Fresnel numbers. We anticipate that the high optical quality will be reduced by refractive nonuniformities that we have not included in the computations.

The temporal advancement of kinetics and optics calculations is treated sequentially for periods Δt. If Δt is not controlled carefully, the relaxation oscillations that result from coupling the laser medium to the radiation field can cause undamped numerical instabilities. We have used a Δt corresponding to half of a photon round trip in the resonator, although this approaches the limit of numerical stability for a 2-m cavity.

VI. CONCLUSIONS

The KrF discharge is attachment dominated for normal Kr concentrations. Two discharge equilibria exist, corresponding to two limiting cases of metastable species loss processes: (a) loss dominated by excimer formation and (b) loss dominated by metastable species ionization. The electron beam is the primary ionization source for the excimer formation equilibrium, whereas the dc electric field is the primary electron source for the ionization equilibrium. The discharge eventually enters the ionization equilibrium, which is stable against fluctuations in the electron density. The discharge is controlled by Kr** ionization at late times. Since the Kr concentration is small, not enough Kr_2^+ can be formed from the Kr^+ to restrain electron density growth by electron-ion dissociative recombination. Thus, discharge current runaway occurs when the F_2 concentration is depleted, so that dissociative attachment no longer balances ionization of Kr**.

The dominant ionization process in the discharge plasma is multistep: ionization of Kr*, Kr**, Ar*, and Ar** by secondary electrons. This introduces power limits because the fractional concentration of metastable species must be kept low ($\sim 10^{-5}$ compared with the ground-state population) to avoid discharge current runaway. But, it also imposes limitations on intrinsic efficiency: the electrons needed to ionize metastable species come from the tail of the electron energy distribution, robbing it of energy needed to excited further Kr* and Ar*. Thus, even if the resulting ion forms KrF* by neutralization followed by displacement, the net loss to the discharage is still approximately 4 eV.

The total gas pressure of a discharge is limited ($\leqslant$3 atm)

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by the fact that the dominant quenching mechanism is a three-body reaction. Quenching dominates interception in virtually all laser experiments performed. At low densities, discharge pumping is inefficient because not enough energy can be loaded in the gas. At high densities, quenching limits enhancement, although high specific energies can be produced.

The major limitation to power extraction is absorption by the molecular ions (particularly Kr_2^+) and the triatomic species (Kr_2F^*, e.g.). The latter, formed through KrF*, is saturable; but the former is not, and hence constitutes the most severse limitation on high-power operation of KrF lasers. Strong pumping produces high electron densities and, consequently, high molecular ion densities.

Since operation at high electron densities is undesirable, electron impact processes such as upper-state mixing by background electrons and superelastic deactivation of KrF* should not pose a fundamental constraint for long-pulse (500 nsec) operation. These two processes were not included in our model, so this statement is predicted on the assumption that the rate constants for these processes do not exceed 10^{-6} cm^3/sec. The cross sections are not available so we cannot compute the rate constants; however, the cross sections would have to be very large for the rate constants to be larger than 10^{-6} cm^3/sec. For an electron density of 1.5×10^{14} cm^{-3} and superelastic and state mixing rate constants of 10^{-6} cm^3/sec these processes are comparable to KrF* loss by spontaneous emission, a loss that is unimportant under well-saturated operation.

The fundamental discharge problem is that the argon and krypton metastable states lie too far from the ground state and too close to the ionization threshold. To make up for collisional losses, discharge enhancement would have to be well in excess of 5; experimental enhancements have not exceeded about 3. Simulation predict sever efficiency penalties for achieving higher enhancement.

The highest intrinsic efficiencies attained experimentally (in volumes of 1 liter or greater) are about 12% in direct electron-beam excitation and about 9% in e-beam-sustained discharge excitation. Specific energy at large volume is limited to about 10 J/liter in the discharge; beam-pumped values are slightly higher at high efficiency, and have reached 17–18 J/liter at somewhat lower ($\sim$7%) efficiency.

These experiments have all been replicated by simulation with accuracies $\approx$10%, well within experimental error.

An extensive computational study of the available parameter regime indicates that the achieved intrinsic efficiencies cannot significantly be improved upon. Specific energies of up to 30 J/liter are theoretically possible; the very best proposed device is shown in Table IV; its efficiency is only 4%. Further, these conditions required uniformity of beam and tailoring of discharge so fine as to be practically unattainable.

We reluctantly conclude that, for KrF, discharge pumping is an unattractive alternative to direct pumping.

A preliminary physical optics analysis shows that injection locking offers no advantage for shortening mode build-up time or improving optical quality.

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